Multi-Stage Fine-Tuning of Pathology Foundation Models With Head-Diverse Ensembling for WBC Classification

Antony Gitau[1,2], Martin Paulson[2], Bjørn-Jostein Singstad[1,2],
Karl Thomas Hjelmervik[1], Ola Marius Lysaker[1], Veralia Gabriela Sanchez[1]

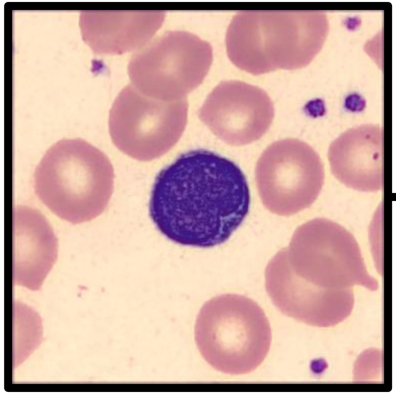
[1]University of South-Eastern Norway,
[2]Vestfold Hospital Trust, Norway

11th April 2026

Universitetet i Sørøst-Norge

SYKEHUSET I VESTFOLD

The challenge is to classify a single-site WBC image

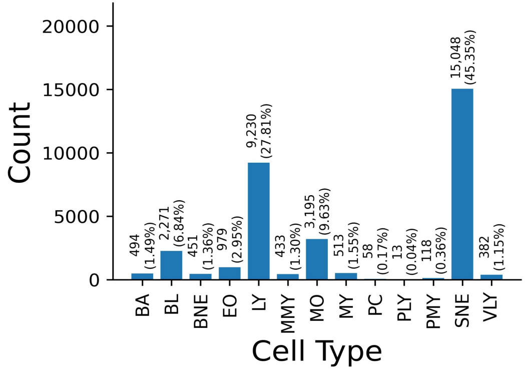


Classification model

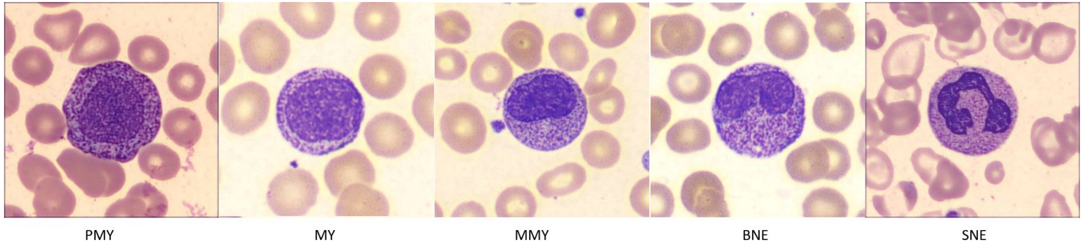
White blood cell classes

- Segmented neutrophil
- **Lymphocyte**
- Monocyte
- Eosinophil
- Basophil
- Variant (atypical) lymphocyte
- Band-form neutrophil
- Metamyelocyte
- Myelocyte
- Promyelocyte
- Blast cell
- Plasma cell
- Prolymphocyte

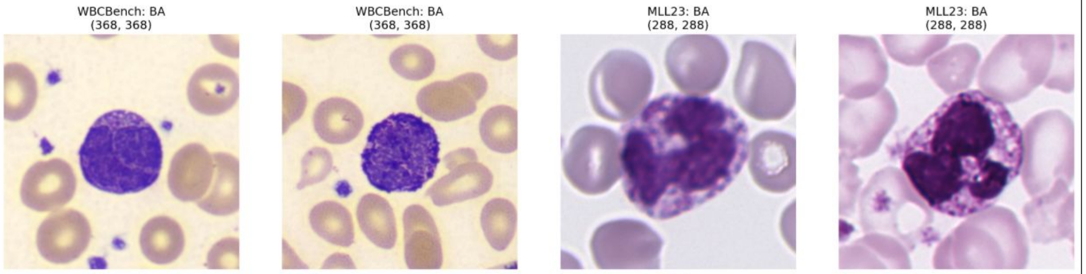
So what make this task hard?



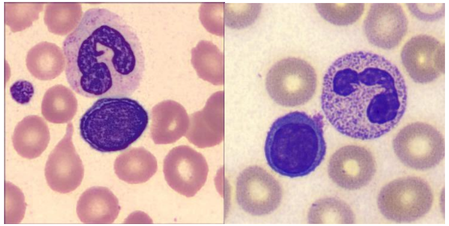
Underrepresented morphologies



Morphology continuum



Boushehri et al., A large expert-annotated single-cell peripheral blood dataset for hematological disease diagnostics, Scientific Data, 2025.



Multi cell images

Variability in imaging conditions

So here is a summary of our contribution

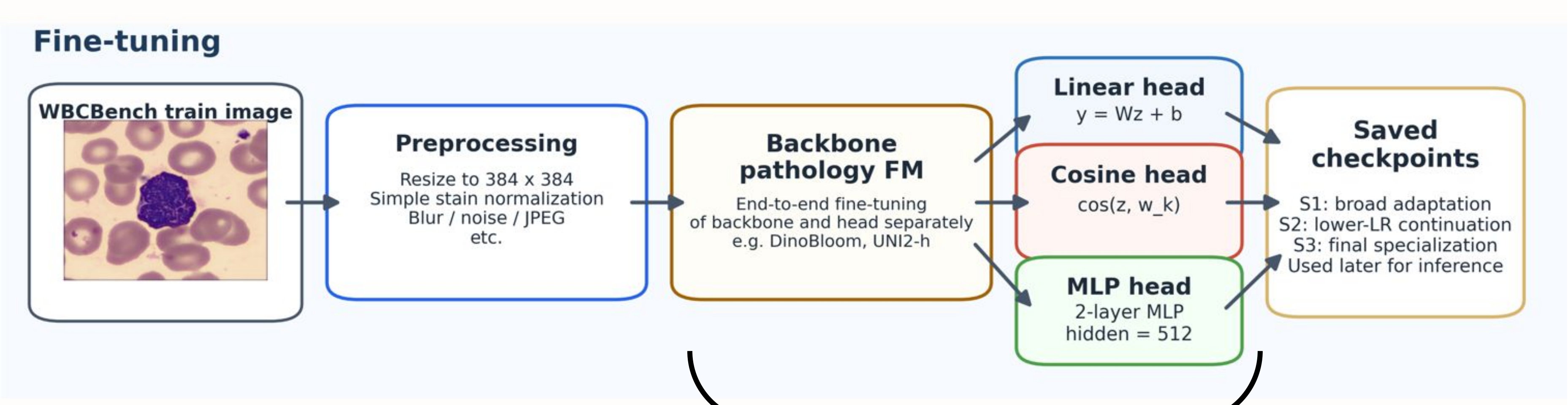
- Present a reproducible fine-tuning strategy for pathology foundation models (PFM).

  (code *https://github.com/Antony-gitau/msfit*)

- Show that the geometry of the classifier head drives class-specific specialization contributing to resolving confusions along morphological continuum.

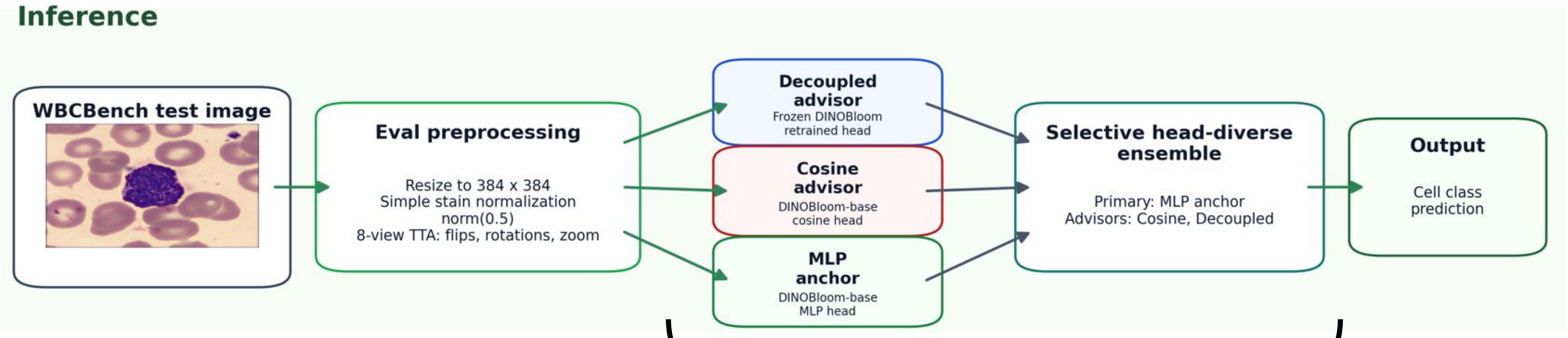
- Analyze the WBCBench dataset, showing that many difficult cases may reflect labeling errors.

Here is our fine-tuning setup for PFM



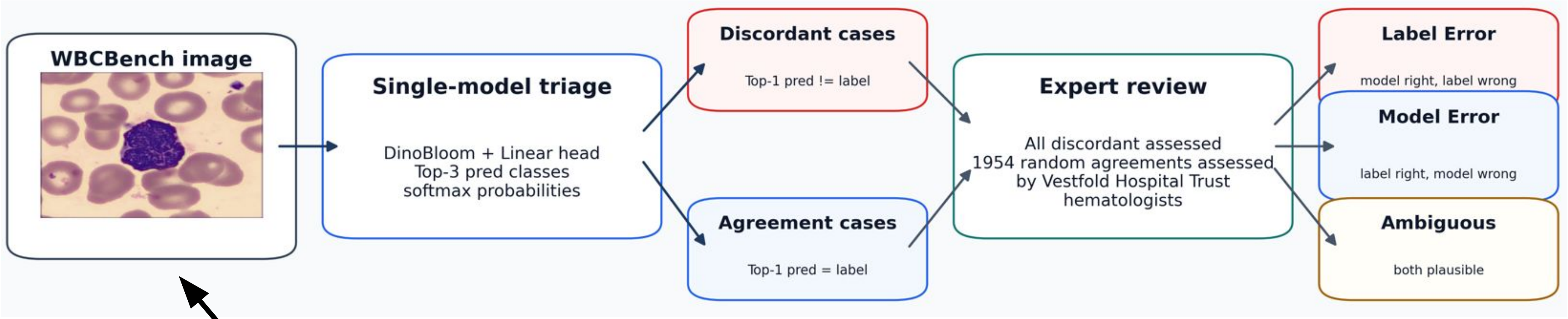
End to end fine-tuning of the PFM backbone and classifier head across 3-staged optimization

And this is the inference setup



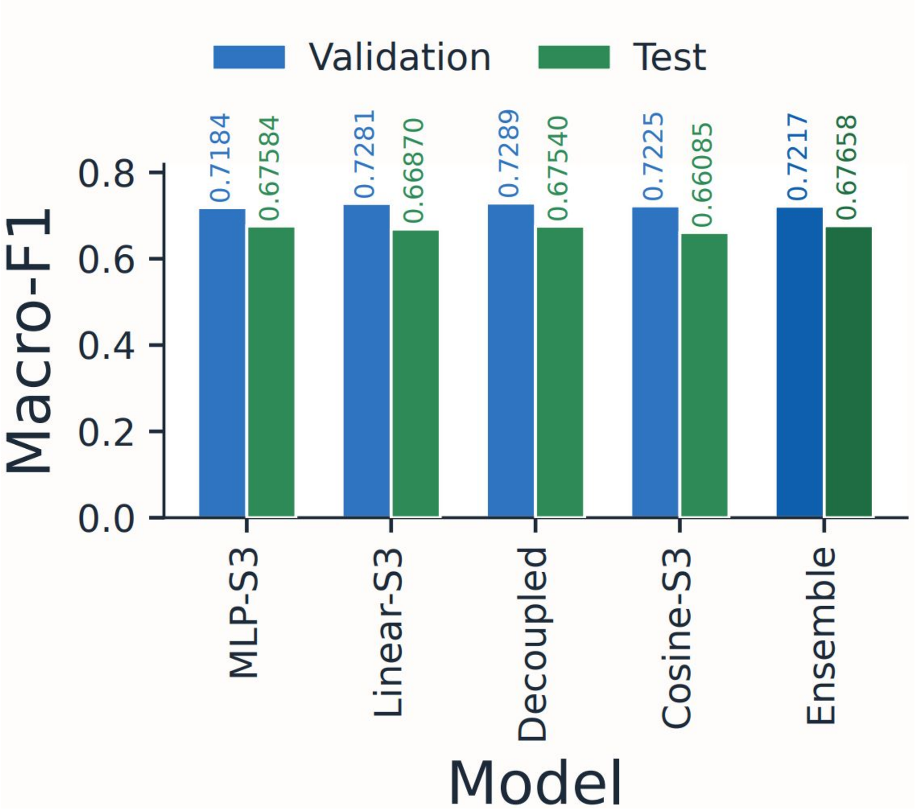
Three separately saved checkpoints forming a diverse head ensemble

And this is how we analyze a subset of the WBCBench



The train and eval sets

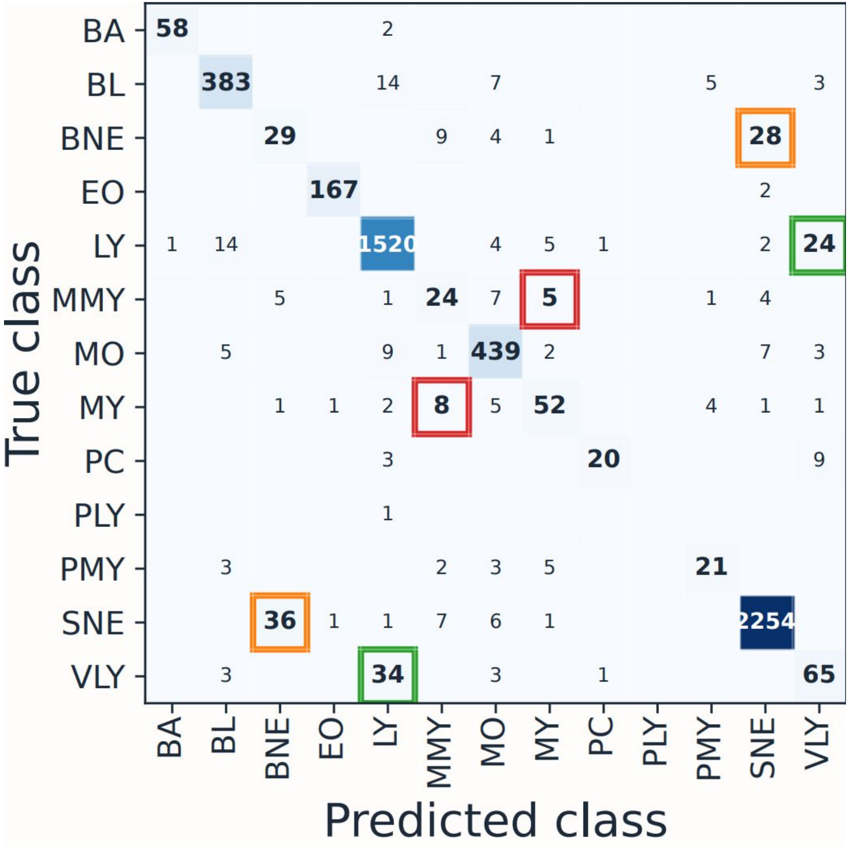
Comparing the performance of some of our models used at inference



Ensemble was our best entry in the challenge

Noticed eval to test drop of ~0.045 on F1 across all models

Confusion occurs between adjacent morphologies

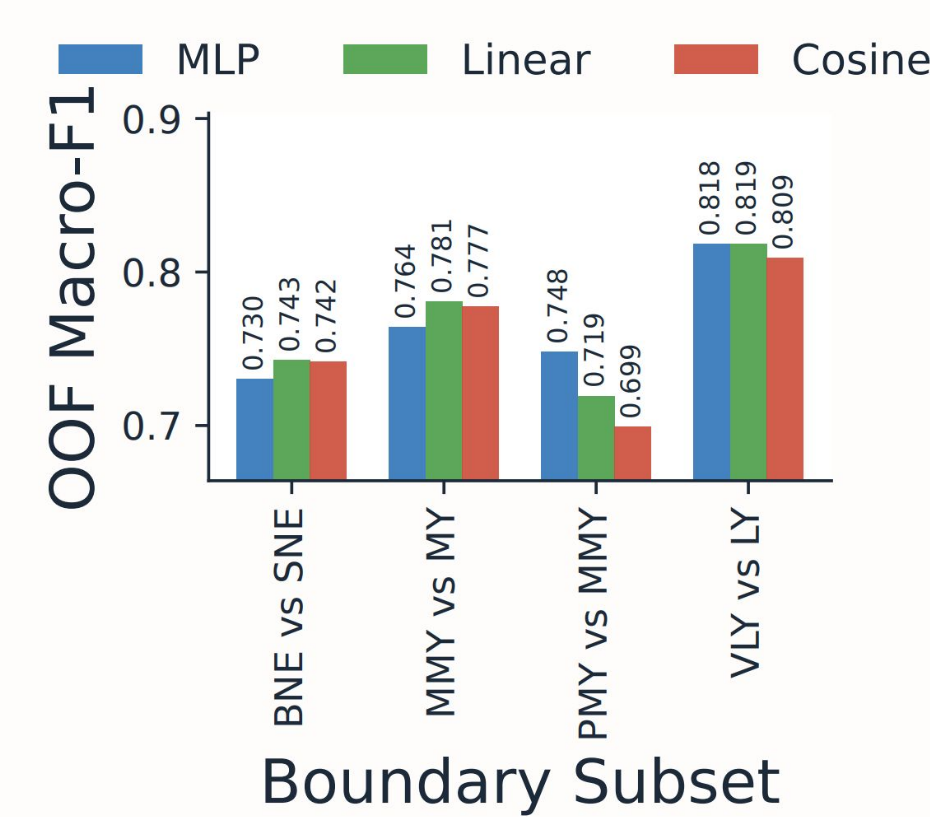


PMY → **MY → MMY** → **BNE → SNE**

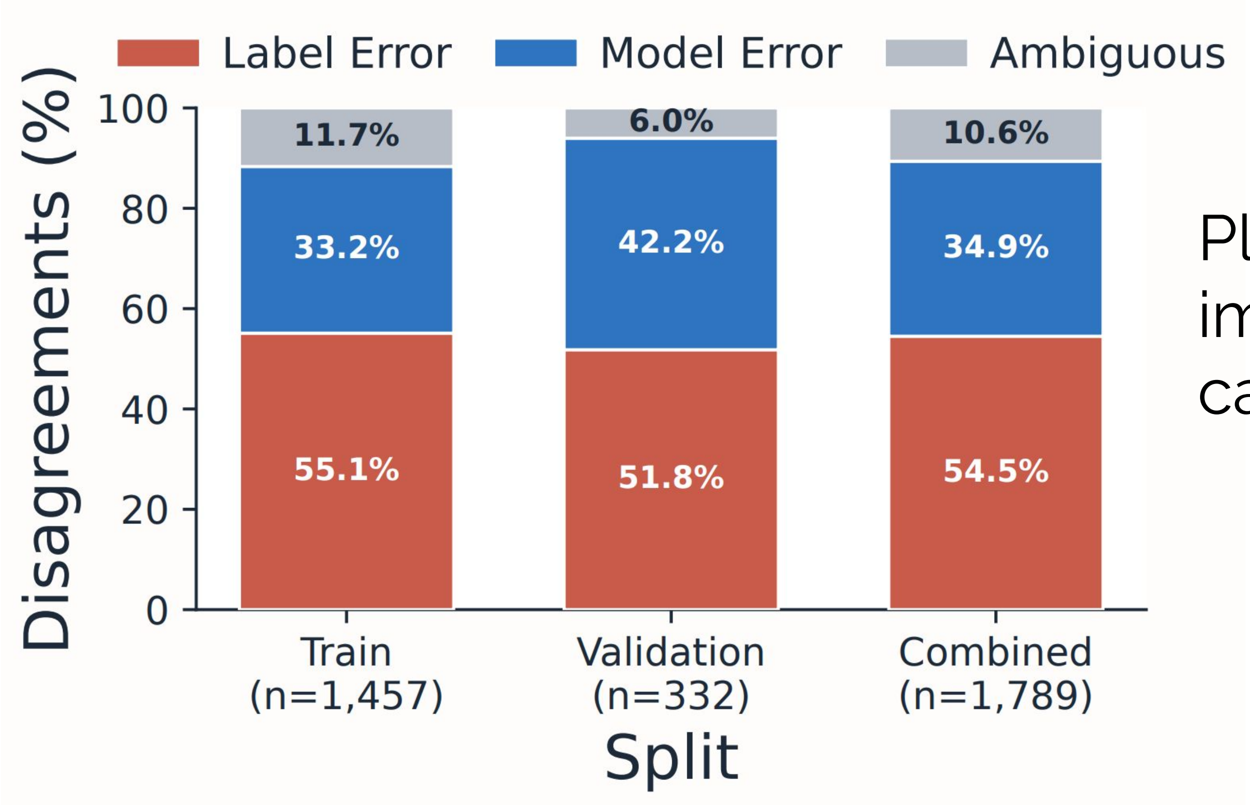
BL → PLY → **LY → VLY** → PC

BA, EO, MO (distinct lineage)

Evaluation shows boundary performance is head-dependent



Among the disagreements, 54.5% were judged as label errors



Plus 8% of the 1,954 images in agreement cases

In a nutshell,

- PFMs are extending to cytology beyond histopathology
  (code https://github.com/Antony-gitau/msfit)

- Classifier head geometry drives performance in fine-grained cell classification.

- Label noise is still non-trivial, motivating need for better annotation tools, benchmarking, and label-robust methods.